

HerHealthEval: Do Language Models Understand Women’s Health Communication Across Languages and Cultures?

Anonymous ACL submission

Abstract

Large language models are increasingly used to answer women’s-health questions, but fluent answers do not establish that a model has correctly understood what a patient is trying to communicate. We present HerHealthEval, a preliminary multilingual evaluation framework and protocol for studying language understanding in women’s-health communication across languages and linguistic registers.

Our controlled evaluation suite covers menstruation, polycystic ovary syndrome (PCOS)/hormonal concerns, and fertility in English, Arabic, and French, pairing clinical, lay, indirect, ambiguous, and emotionally concerned expressions with predefined targets for concern category, risk level, recommended action, and clarification need. The protocol requires native-speaker validation of Arabic and French for meaning preservation, naturalness, and register before cross-lingual evaluation or adaptation, while evaluation cases remain separated from leakage-controlled adaptation data. The experimental design centers Qwen3.5-9B and specifies a general-purpose baseline, English domain adaptation, and subsequent Arabic/French adaptation, evaluated through misunderstanding, under-triage, clarification, unsafe-response, and cross-style/cross-language consistency measures. Rather than claiming comprehensive clinical coverage, this preliminary study establishes a failure-oriented protocol for testing women’s-health language understanding beyond answer generation.

1 Introduction

The central question here is not whether a language model can produce a plausible answer to a women’s-health question, but whether it first understands what the patient is trying to communicate. The same concern may be stated clinically (“I have irregular menstrual bleeding”), colloquially (“my period is all over the place”), or indirectly

(“my monthly problem has changed”). These expressions point to the same underlying concern but differ in explicitness, emotion, and cultural register; misreading that variation can lead a model to confuse the concern, understate urgency, offer false reassurance, or answer when it should first ask for missing information.

Most women’s-health LLM studies instead evaluate the generated answer rather than whether the input was understood, and multilingual health QA work shows the same question can elicit inconsistent information across languages. Together, these gaps motivate a controlled evaluation of *interpretation* rather than another answer-generation study; §2 reviews this literature in full.

We present **HerHealthEval**, an initial investigation of language understanding in multilingual women’s-health communication. The evaluation suite spans menstruation, PCOS/hormonal concerns, and fertility, holding each underlying case fixed while varying canonical, clinical, layperson, indirect/culturally sensitive, ambiguous, and emotionally concerned expression. Matched English cases are localized into Arabic and French under a native-speaker-approved protocol before entering cross-lingual evaluation or adaptation. Each item has predefined targets for concern category, risk level, recommended action, and clarification need, and the evaluation lineage is kept separate from the adaptation corpus by source- and paraphrase-family exclusion.

We organize the study around three research questions:

- RQ1:** How robust are LLM-based clinical reasoning systems to variation in linguistic register when interpreting semantically equivalent women’s health concerns?
- RQ2:** Do multilingual LLMs exhibit systematic differences in interpretation, triage cali-

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bration, and clarification behavior across English, Arabic, and French?

3. **RQ3:** What is the impact of domain-specific and multilingual adaptation on linguistic robustness, uncertainty handling, and clarification generation in women’s health symptom understanding?

The contributions align with these questions: (1) women’s-health interaction framed as a controlled language-understanding problem across register; (2) a multilingual evaluation protocol with matched language variants, a native-speaker validation gate, and leakage-separated adaptation data; and (3) a reproducible, failure-oriented comparison of general-purpose and sequentially adapted models on misunderstanding, triage, clarification, safety, and consistency. These contributions are deliberately preliminary: an evaluation protocol and early failure patterns, not comprehensive coverage or clinical readiness.

2 Related Work

We cluster the closest work into three lines: women’s-health answer evaluation, multilingual and sociocultural language understanding, and evaluation design for ambiguity and safety.

2.1 Women’s-Health Answer Evaluation

Women’s-health LLM research primarily asks whether generated answers are accurate, readable, empathetic, or safe (Recker et al., 2025; de Vries et al., 2025). Adhikary et al. (2025) introduce MenstLLaMA, one of the closest domain-adaptation systems, fine-tuning LLaMA-3-8B on menstrual-health QA data and evaluating educational response quality in English. Mughal et al. (2025) develop Mai, a menstrual-health dialogue system in English and Roman Urdu motivated by stigma and sociocultural barriers, but evaluated through automatic metrics and user ratings rather than matched interpretation tests. For PCOS, Devranoglu et al. (2024) evaluate ChatGPT on clinician-authored questions and Graça et al. (2026) run an international blinded survey of health-professional ratings; a systematic review of 80 PCOS AI studies finds the broader field still dominated by imaging and structured predictive modeling (Ghaderzadeh et al., 2025), so conversational LLM evaluation is a small slice of that literature. For fertility, Chervenak et al. (2023) compare ChatGPT counseling with public-health

sources, and Gély et al. (2026) compare LLM answers to infertility FAQs against specialist answers. Maurya et al. (2026) broaden coverage through WHBench, an expert-crafted, safety-aware rubric. These studies motivate domain-specific adaptation and careful response evaluation but treat the input question as fixed; none tests whether the same concern is interpreted consistently as its register or language changes.

2.2 Multilingual and Sociocultural Understanding

Jin et al. (2024) introduce X LingEval to assess correctness, consistency, and verifiability on health-care questions across English, Spanish, Chinese, and Hindi, finding substantial cross-language disparities; subsequent work likewise reports inconsistent information across languages for equivalent questions (Schlicht et al., 2025). Foundational cross-cultural NLP work distinguishes linguistic diversity from cultural variation, identifying linguistic form, common ground, topic relevance, and communicative objectives as separate dimensions (Hershcovich et al., 2022; Adilazuarda et al., 2024). In reproductive health, participatory work with underserved women shows culturally appropriate responses require local context beyond literal translation (Deva et al., 2025), while Western-centric rubrics can underrate responses trained annotators judge appropriate (Dey et al., 2025). LUHME work identifies related problems: models compress cultural variation (Mohammadi et al., 2025), adaptation can fail when sociocultural context is ignored (Freitas and Cardoso, 2025), and dialogue systems need mechanisms for common ground (Anikina et al., 2025). Concept-level terminological precision, not surface fluency, matters most where domain mismatches can alter meaning, as in health-care (Heinisch, 2025). Cross-register transfer work further motivates holding semantic content constant when measuring sensitivity to linguistic form (Nicholls and Alperin, 2025). Our setting brings these concerns together in matched English, Arabic, and French women’s-health descriptions rather than testing general medical knowledge alone.

2.3 Gap and Evaluation Requirements

The resulting gap is a framework that separates understanding the input from generating an appealing answer. Clarification research treats detecting the need to ask, selecting an informative question, and responding after clarification as distinct capabilities

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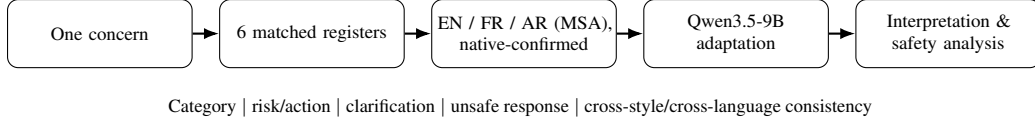


Figure 1: Conceptual overview. Clinical content is held constant while linguistic form and language vary, separating interpretation failures from changes in the underlying case.

that current models still struggle with (Testoni and Fernández, 2024; Zhang and Choi, 2025); structured, stateful dialogue that elicits missing information turn by turn is one practical direction (De-marco et al., 2024). Such a framework must (i) hold the underlying concern constant across register and language, (ii) score interpretation, risk/action, and clarification need separately, (iii) expose under-triage and unsafe advice rather than rely on aggregate accuracy, and (iv) prevent overlap between evaluation and adaptation data. Native-speaker review targets semantic and register preservation in Arabic and French, while source- and paraphrase-family exclusions separate evaluation from training. The blinded clinical-rating designs used in PCOS and fertility studies motivate independent review of response quality (Graça et al., 2026; Gély et al., 2026), but our primary measures remain tied to predefined item labels.

LLM judges show only moderate agreement and systematic rubric biases (Donati et al., 2025), so rubric-scored cultural sensitivity, helpfulness, clarity, and safety are reported separately from deterministic label metrics; pairing an LLM judge with a human-verification step is one proposed mitigation (Grießhaber et al., 2025).

3 Methods

This is an initial-exploration study: the protocol below describes a general procedure rather than a finished, fixed instrument. Design and construction are described qualitatively; exact item and split counts are intentionally omitted (§5.1).

3.1 Design philosophy: matched cases, controlled register

The evaluation is built around one idea: express a single underlying women’s-health concern in several ways a real patient might, while keeping the clinical facts fixed. Each case is written once in a canonical, evidence-anchored form and rewritten into five further *registers* — clinical, lay, indirect/culturally hedged, ambiguous/under-specified, and emotionally concerned — so that any change

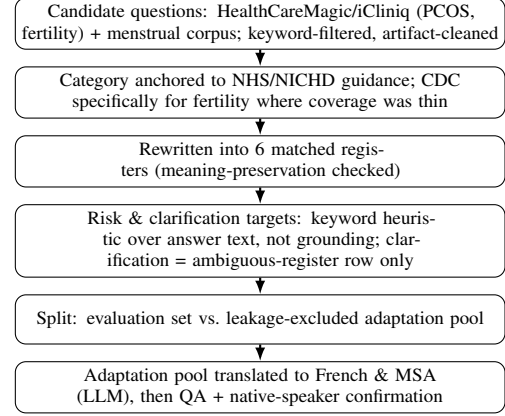


Figure 2: Data construction pipeline (schematic).

in predicted concern category, risk level, or clarification need can be attributed to *how* the case is phrased (Figure 1, Figure 2). Every case carries a predefined target for category, risk, action, and clarification need, so scoring never compares free text directly. For example, a case about heavy menstrual bleeding might appear as a clinical chart-note question in its *clinical* register and as a short, worried, under-specified message in its *ambiguous* register; only the ambiguous variant targets clarification, since it is specifically constructed to omit a detail a clinician would need to ask for (§3.4).

3.2 Seed curation

Candidate patient-authored questions for PCOS/hormonal and fertility content were drawn from the HealthCareMagic-100k and iCliniq-10k patient–doctor dialogue datasets (Li et al., 2023; Li, 2023), accessed via a Hugging Face repackaging (Malikeh1375, 2023), supplemented by additional patient-question corpora for menstrual-health content. Candidates were filtered by keyword for the three categories; cleaned of a recurring text-substitution artifact present in a subset of ChatDoctor-derived answers; filtered for authentic first-person framing against near-duplicate, myth-only, or off-topic content; and balanced across categories. Where a case’s claim mapped to a public-health-authority page, its category was anchored to that source — NHS and NICHD

Study	Primary target	Languages	Matched variants	register	Explicit triage / clarification targets
MenstLLaMA (Adhikary et al., 2025)	Educational answer generation	English	No		No
Mai (Mughal et al., 2025)	Menstrual-health dialogue generation	English, Roman Urdu	No		No
WHBench (Maurya et al., 2026)	Expert-scored clinical responses	English	No		Safety rubric; no clarification target
HerHealthEval (this work)	Symptom-language understanding	English (FR/AR: adaptation data)	Yes		Yes

Table 1: Positioning against the closest women’s-health LLM systems. French and Arabic here refer to the adaptation-data languages; the scored evaluation set is English (§5.1).

guidance broadly, with CDC guidance used specifically for fertility content where NHS/NICHD coverage was thin; unmapped claims were flagged rather than assigned an invented category. This anchoring pass is deterministic and rule-based, not model-generated.

3.3 Register construction

Each canonical text is rewritten into the remaining five registers under a meaning-preservation constraint: vocabulary, directness, hedging, and affect may vary, but no clinical claim may be added or removed. An early template-based rewriter collapsed distinct cases onto near-identical clinical-register text and was replaced with a pass that verifies distinctness and meaning preservation. Every register variant is flagged for human spot-check.

3.4 Gold risk and clarification targets

An early approach tried to ground the risk-level target in the same evidence pages used for category anchoring; this was abandoned once it produced a single, non-discriminating label for effectively every case. The risk target used here instead comes from a disclosed, deterministic keyword rule over the case’s answer text, computed once per case so all of its register variants share one ground-truth risk label. The clarification target is register-determined by construction: *yes* exactly for a case’s ambiguous-register row, *no* otherwise. Both targets are explicitly *silver* — heuristic, not clinician-adjudicated — and used identically for the evaluation set and the adaptation data.

3.5 Multilingual localization

Matched English cases are localized into French and Modern Standard Arabic (MSA) rather than a regional dialect, so terminology generalizes across Arabic-speaking audiences. Both corpora were produced with the same LLM translation pipeline

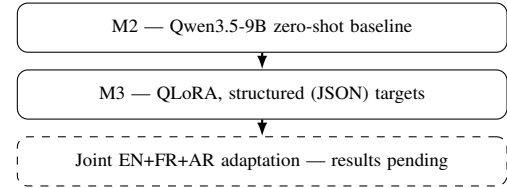


Figure 3: Model adaptation lineage (schematic).

(OpenAI’s Responses API, gpt-5.6-sol, low reasoning effort, structured outputs), preserving each row’s register label rather than flattening every register into one translation. Output then passes an automated linguistic screen (structural integrity, encoding, source–target alignment) and native-speaker confirmation of meaning and register; for Arabic this is a professional Arabic↔English medical translator working in MSA, checking terminology and grammar (including gender agreement in patient-directed phrasing).

3.6 Evaluation/adaptation separation

Evaluation items are held out from every adaptation corpus by two keys: exact (case-insensitive) question match, and shared source-provenance identity, which also removes paraphrase-family overlap. The rule applies uniformly across English, French, and Arabic, and train/validation duplicates are resolved so the same case cannot inform both.

3.7 Models and adaptation protocol

All experiments use Qwen3.5-9B. We report a zero-shot baseline (M2): the base model prompted with a structured triage schema for category, risk, action, and clarification. We then adapt with QLoRA on the leakage-controlled data above, wrapping targets in the exact structured schema the evaluation items score against (M3); alternative target formats were explored during development and are not reported here. A joint English+French+Arabic run follows the same protocol and is scored the same

way (Figure 3); results are reported separately once available.

3.8 Metrics

We score four predefined-label dimensions rather than free-text quality: *category accuracy*; *risk accuracy*, reported against the majority-class baseline since labels are skewed; *under-/over-triage rate*, the safety-relevant direction of risk error; and *clarification recall/specificity*. We also report *parse compliance* and a *cross-style consistency* rate (the fraction of a case’s register variants given the same label); an analogous cross-language rate applies once a scored multilingual item set exists. A model-self-reported `unsafe_response` flag is logged but treated as weak, non-validated signal. Paired differences between the baseline and each recipe use McNemar’s test on matched per-item correctness.

4 Results

We report results on the English evaluation set for two Qwen3.5-9B variants: the zero-shot baseline and our QLoRA adaptation on structured (JSON) targets. Results for French and Arabic will be reported separately once available. Gold labels are skewed, so we foreground majority-class baselines alongside accuracy-style numbers and treat per-class recall, under-triage, and clarification recall as the honest headlines. The risk majority-class baseline is 0.644 and the clarification majority-class baseline is 0.833.

4.1 Zero-shot baseline (M2)

Table 2 reports the zero-shot baseline (M2), an instruction-tuned Qwen3.5-9B prompted with our structured triage schema. M2 parses at 100% and reaches 0.667 risk accuracy, only marginally above the 0.644 majority baseline. The safety-critical weakness is *under-triage*: of 174 items whose gold risk is see-doctor, M2 routes 71.8% to routine (per-class recall 0.264, versus 0.897 for routine). M2 retains strong clarification-seeking, however: recall 0.856 on the 90 items requiring clarification, at 0.862 specificity. Category accuracy is 0.539, with fertility weakest (recall 0.467). The model gives a consistent risk label to only 64.4% of case groups across their six phrasings, and a consistent category to just 15.6% — direct evidence that surface communication style, not clinical content, drives much of the interpretation variance (RQ1, §5.1).

Metric	M2	M3	Base
parse_ok	1.000	0.998	—
risk accuracy	0.667	0.623	0.644
under-triage ↓	0.718	0.647	—
clarif. recall	0.856	0.044	—
category accuracy	0.539	0.549	—
consistency (risk)	0.644	0.433	—
consistency (cat.)	0.156	0.344	—

Table 2: Results on the English evaluation set ($n = 540$). M2 = zero-shot baseline; M3 = QLoRA fine-tune on structured (JSON) targets. “Base” = majority-class baseline. Arrows: lower is better for under-triage. Best per row in **bold**. Consistency = fraction of case groups given the same label across all six styles.

4.2 Fine-tuning reduces under-triage but erases clarification-seeking

We fine-tune Qwen3.5-9B with QLoRA on leakage-controlled adaptation data drawn from the same case pool as the evaluation set but disjoint from it, using targets wrapped in the exact structured schema the evaluation items score against.

M3 improves parse compliance (0.998) and reduces under-triage below the zero-shot baseline (0.718 to 0.647), and more than doubles cross-style category consistency (0.156 to 0.344). But its risk accuracy is significantly worse than the zero-shot baseline (0.623 vs. 0.667; McNemar’s $p=0.03$, favoring M2), and clarification recall — 0.856 for the zero-shot baseline on the 90 items that require it — collapses to 0.044: fine-tuning all but eliminates clarification-seeking. Category ($p=0.71$) and clarification ($p=0.30$) correctness are not significantly different from M2; parse compliance is statistically unchanged ($p=1.0$).

5 Discussion

Two results stand out. General-purpose Qwen3.5-9B under-triages the majority of care-warranting cases and interprets the same case differently depending only on phrasing (RQ1, §5.1). Adaptation is a double-edged tool: it reduces under-triage and stabilizes category interpretation, but at some cost to risk accuracy, and it erases clarification-seeking almost entirely. This suggests clarification on ambiguous input is not a skill that answer-style supervised fine-tuning teaches by example; recovering it likely needs objectives that explicitly reward abstention or asking — preference optimization, or a dedicated clarification-vs-answer decision head — rather than more demonstrations. The practical lesson for safety-relevant domains: evaluate adaptation on the structured behaviors that matter at

deployment, triage *and* clarification, not answer quality alone, since a model can improve on one while losing another.

5.1 Preliminary answers to the research questions

We revisit the three research questions against the evidence available at this stage; all are answered only as far as current results allow.

RQ1 (register robustness). *Partially answered, English only.* Holding a case fixed and varying only its phrasing changes the predicted risk label for 35.6% of cases and the predicted category for 84.4% (cross-style consistency 0.644 and 0.156). Adaptation partly redistributes this: M3 more than doubles category consistency (0.156 to 0.344) but lowers risk consistency (0.644 to 0.433), so robustness gains are not uniform across target types.

RQ2 (cross-lingual differences). *Not yet answerable.* Results for Arabic and French will be reported separately once available; we do not extrapolate the English-only findings above to other languages.

RQ3 (adaptation impact). *Partially answered, English-only adaptation.* Domain adaptation reduces under-triage (0.718 to 0.647) and improves cross-style consistency, but at a cost to risk accuracy and with clarification recall collapsing to near zero. Results for the multilingual (English+French+Arabic) adaptation stage will be reported separately once available.

Limitations

Results reported here are on the English evaluation set; results for French and Arabic will follow separately. Gold labels are skewed toward routine risk and no-clarification, which is why we foreground majority baselines, per-class recall, and under-triage over raw accuracy. Risk and clarification targets are a disclosed heuristic rather than clinician adjudication, applied identically to the evaluation set and the adaptation data (§3.4). The `unsafe_response` flag is model-self-reported, not an independently validated safety measure. The Arabic adaptation data uses Modern Standard Arabic only. Since Qwen3.5-9B is itself a multilingual model with pretrained exposure to Arabic dialects, this is a gap in our controlled data coverage rather than evidence that the model cannot handle dialectal input — but we have not directly measured di-

alectal performance, and extending evaluation to dialectal Arabic (e.g., Egyptian, Gulf, Levantine) is a natural direction for future work. This initial study covers three symptom categories, three languages, and a single 9B model family, so generalization to other conditions, cultures, care settings, and model scales is untested. Arabic and French adaptation text was produced by LLM machine translation and confirmed by native-speaker/professional review for meaning and register (§3); this validates translation quality, not the clinical accuracy of gold targets. Model outputs are evidence of linguistic understanding in a controlled research setting, not medical advice or proof of clinical-deployment readiness.

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